# Real-Time Lens Blur Effects and Focus Control

IG3D - MU4IN602 – Research paper contribution abstract

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**Abstract:**

Our project implements and extends the layered lens blur rendering method introduced in *Real-Time Lens Blur Effects and Focus Control* by Lee, Eisemann, and Seidel (2010), adapting it for use within a Blender compositing workflow. The original paper presents a real-time depth-of-field simulation that uses a two-stage depth peeling strategy to separate foreground and background occlusions, enabling more accurate focus transitions. It also incorporates aperture-shaped blur kernels (Bokeh) and simulates optical aberrations, such as spherical aberration, chromatic dispersion, and field curvature, to achieve photorealistic effects while maintaining computational efficiency.

In our work, we faithfully implement the core depth-of-field model from the original paper, utilizing depth peeling and layered compositing. However, we adapt the method to Blender by developing a custom node-based compositor using Blender's Python API. Unlike the original approach, which relies on proprietary shaders or hardware techniques, our method mimics the dual depth peeling structure with Z-buffer sorting, object-based depth masks, and selective Bokeh convolution kernels, offering more flexibility in controlling blur intensity compared to the original single-pass approach.

Lee et al.'s research focuses on efficiently simulating depth-of-field effects with a two-stage depth peeling strategy and real-time blur kernels, enabling realistic bokeh and optical aberrations without sacrificing performance. Our implementation extends this approach by introducing modularity through adjustable parameters for focal depth and transition smoothness.

We mask regions based on their depth from the camera, creating multiple layers for varying distances in the scene. Each layer is processed with different blur levels, influenced by its distance from the focal plane. The circle of confusion (COC) is calculated per object, influencing

the blur intensity for a more realistic depth-of-field simulation. Additionally, chromatic splitting is applied to simulate chromatic aberration.

This technique has been applied to a stylized urban animation sequence, where focus transitions and lens artifacts enhance both artistic expression and technical accuracy. **Figure 1** demonstrates the layered blur effect with occlusion handling, showing front layers greatly blurred due to proximity to the lens and far background layers greatly blurred due to their distance. **Figure 2** highlights the chromatic aberration effect, with accentuated color fringing around out-of-focus areas, further enhancing the photorealism of the scene.

***Figure 1. Annotated layers on a blurred frame (non-restricted to 3 layers)***

***Figure 2. Blurred frame with chromatic aberration***

**Reference:**

Lee, S., Eisemann, E., & Seidel, H.-P. (2010). Real-time lens blur effects and focus control. ACM Transactions on Graphics (Proceedings of SIGGRAPH), 29(4), Article 65. https://doi.org/10.1145/1778765.1778802